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Distributional effects of carbon pricing: An analysis of income-based versus expenditure-based approaches

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ABSTRACT

As nations strive for a just transition to net-zero carbon emissions, the equitable distribution of carbon pricing's impact is of significant importance. Carbon pricing, a strategy recognized for its potentially regressive nature, tends to disproportionately burden lower-income households compared to wealthier counterparts. This study examined how the choice of income or expenditure as metrics for assessing carbon burdens and ranking household economic status influences these distributional effects. Our research, employing input-output models and the Taiwan 2021 Survey of Family Income and Expenditure, compared these two metrics. The findings revealed that income-based assessments would amplify the regressive impact of carbon pricing. Conversely, using expenditure as the base for carbon burden exhibited markedly reduced regressive effects. Moreover, the study highlighted the role of employment structures in determining the equity of ranking households by income or expenditure. The stability of distributional outcomes, when employing expenditure as the assessment base, suggested its superiority as a metric unaffected by ranking methodologies. Additionally, our analysis indicated that including gasoline in carbon pricing policies could counteract the regressive nature of such policies. This aligned with a just transition by potentially distributing the burden more evenly across socio-economic groups. By providing insights into the distributional effects of carbon pricing, this study contributed to the global pursuit of net-zero emissions while adhering to the principles of social equity and justice.

1. Introduction

In recent years, carbon pricing has emerged as a principal policy instrument for decarbonization. As of April 2023, a report from the World Bank indicated that 73 carbon pricing systems are in operation globally, covering approximately 23% of total global greenhouse gas (GHG) emissions (World Bank, 2023). Concurrently, to counteract carbon leakage from imported goods, international carbon tariffs such as Europe's Carbon Border Adjustment Mechanism (CBAM) have been instituted. While carbon pricing stands out as the prominent market-based tool to mitigate the climate crisis, it inevitably ignites debates over its implications for fairness and distributive justice. Implementing environmental policies can have uneven economic consequences, including transition costs, increased product prices passed on to consumers, benefit allocation, and shifts in property values due to these changes (Fullerton, 2011). The Intergovernmental Panel on Climate Change's Sixth Assessment Report (IPCC AR6) also underlined the potential for poorly adopted environmental policies to disproportionately burden vulnerable populations, thereby amplifying inequalities (Pörtner et al., 2022).

A growing body of research is delving into the equity implications of carbon pricing. The redistribution of carbon pricing revenues has garnered extensive debate, primarily stemming from fears that such redistribution might accentuate wealth disparities. This could be through a direct impact on household incomes or an indirect effect on commodity prices (Ampudia et al., 2018). Consequently, carbon pricing can potentially erode the purchasing power of households and consumers. A critical concern is the possibility of regressive distributional outcomes, where the burden of carbon pricing disproportionately affects economically vulnerable households. This regressive impact means that lower-income groups may bear a greater burden relative to their wealthier counterparts, leading to inequitable distributional

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consequences.

An examination of distributional effects across countries illuminated key determinants of carbon pricing's equitable outcomes. A comparative study spanning eight Asian nations discerned that those with higher average incomes tended to witness more profound economic setbacks for their lower-income households due to carbon pricing, thus culminating in regressive consequences (Steckel et al., 2021). This study attributed such outcomes to the regressive nature of electricity consumption in higher-income countries. Another comprehensive analysis across 87 low and middle-income countries worldwide emphasized that an inherently regressive energy sector amplifies the likelihood of regressive carbon pricing effects by a staggering 95% (Dorband et al., 2019). This finding emphasized the pivotal role of the energy sector – which was both carbon-intensive and indispensable – in shaping the distributional outcomes of carbon pricing.

On a country-specific level, scholars have conducted comprehensive assessments of the distributional effects of carbon pricing, especially among different household income groups (Köppl and Schratzenstaller, 2023; Ohlendorf et al., 2021; Wang et al., 2016). For example, Wang et al. categorized the methodological approaches of relevant studies and noted the prevalent use of Input-Output (IO) and Computable General Equilibrium (CGE) models, frequently in conjunction with household income and expenditure surveys (Wang et al., 2016). Ohlendorf et al. on the other hand, emphasized the importance of transportation fuels and national subsidies (Ohlendorf et al., 2021). Both studies concurred that distributional outcomes in developed countries are predominantly regressive, albeit with some exceptions that are neutral or even slightly progressive. Besides variations between countries, the chosen research methodology also substantially affects these outcomes. There is a difference between evaluating carbon burdens using total expenditure versus total income. Köppl and Schratzenstaller point out that evaluation grounded in expenditure often unveils less pronounced regressivity (Köppl and Schratzenstaller, 2023). Most of the existing literature, however, has typically concentrated on employing either income or expenses as metrics for estimating carbon impact.

For a holistic assessment of distributional effects, it is crucial to examine not only the absolute carbon expenditures but also their percentage relative to the total household income or expenditure. The choice between evaluating based on total income or total expenditure can significantly influence the outcomes. Table 1 catalogs literature detailing the distributional effects of carbon pricing across countries. Studies such as those from the United Kingdom (Feng et al., 2010), Ireland (Verde and Tol, 2009), Netherlands (Kerkhof et al., 2008), United States (Grainger and Kolstad, 2010), and France (Berry, 2019) employed total income as their metric, consistently indicating regressive outcomes. In contrast, research in China (Jiang and Shao, 2014; Wang et al., 2019; Yan and Yang, 2021), Italy (Tiezzi, 2005), and Spanish (Labandeira and Labeaga, 1999) using total expenditure as the evaluation measure showed proportional or mildly progressive distributional effects. A few investigations, like those in Denmark (Wier et al., 2005), the United States (Mathur and Morris, 2014), and Canada (Beugin et al., 2016), compared both methods, revealing that the regressive nature of outcomes diminishes considerably when anchoring evaluations in total expenditure, occasionally even shifting towards a mildly progressive trend. A recurrent theme across these studies posited that income inequality is a primary factor driving the regressive distributional effects of carbon pricing (Andersson and Atkinson, 2020). Thus, the choice between income and expenditure as an evaluation basis profoundly shapes distributional results.

In the existing literature, the distributional effects of energy taxes have been explored in the OECD study conducted by Flues and Thomas, who used both income and expenditure as evaluation bases (Flues and Thomas, 2015). While the merits of using income include its direct comparability with other income-based taxes and the fact that compensation is usually calculated based on income, the report argued for using expenditure as the basis. This is because many low-income households may exhibit higher consumption levels due to past savings or borrowing against anticipated income; some rich households may also be low on income but high on wealth. Using income could thus overstate regressive impacts. By adopting the expenditure basis, the influence of borrowing and savings is effectively removed, offering a more reliable measure of the current standard of living. This perspective closely aligns with the Permanent Income Hypothesis (Friedman, 1957), which posits that households adjust their current consumption based on their anticipated future income, making annual expenditures a better representation of lifetime well-being. Beyond the debate over income versus expenditure as determinants of tax burden, there is another pivotal issue of how best to define the social class of households. Specifically, this issue explored whether income or expenditure is more appropriate for ranking households. Pizer and Sexton's research highlighted the significant impact of the ranking method on distributional outcomes (Pizer and Sexton, 2019). This study, rooted in the Permanent Income Hypothesis, advocated expenditure ranking as the superior approach. However, another OECD study on the distributional effects of

Table 1
Existing literature on the distributional effects of carbon pricing.

Author(s)	Country	Methodology	Criteria	Result
Feng et al. (Feng et al., 2010)	United Kingdom	Input-output model	Income	Regressive
Verde and Tol (Verde and Tol, 2009)	Ireland	Input-output model	Income	Regressive
Kerkhof et al. (Kerkhof et al., 2008)	The Netherlands	Input-output model	Income	Regressive
Grainger and Kolstad (Grainger and Kolstad, 2010)	United States	Input-output model	Income	Regressive
Berry (Berry, 2019)	France	Microsimulation	Income	Regressive
Jiang and Shao (Jiang and Shao, 2014)	China	Input-output model	Expenditure	Regressive
Wang et al. (Wang et al., 2019)	China	Input-output model	Expenditure	Weak regressive[a.]
Labandeira and Labeaga (Labandeira and Labeaga, 1999)	Spanish	Input-output model, Microsimulation	Expenditure	Proportional
Yan and Yang (Yan and Yang, 2021)	China	Input-output model	Expenditure	Progressive
Tiezzi (Tiezzi, 2005)	Italian	Ideal demand system	Expenditure	Progressive
Wier et al. (Wier et al., 2005)	Denmark	Input-output model	Income and Expenditure	Regressive in income, Proportional in expenditure
Mathur and Morris (Mathur and Morris, 2014)	United States	Input-output model	Income and Expenditure	Regressive in income, Less regressive in expenditure
Beugin et al. (Beugin et al., 2016)	Canada	Input-output model	Income and Expenditure	Regressive in income, proportional or progressive in expenditure

a. According to Wang et al. the overall regressivity of carbon pricing in China is significantly lower than that of direct carbon emission pricing, exhibiting only weak regressivity.

consumption taxes presented a divergent perspective on this matter (Brys, 2014). It argued that the unique economic circumstances of each household merit different and fairer ranking methods, as shown in Table 2.

Table 2 synthesizes the ranking recommendations provided by the OECD study for various economic situations (Brys, 2014). The essence of this discussion centered on the importance of using criteria that accurately reflect a household's lifetime economic standing. For example, for households that are lifetimes-rich but currently have low income and high expenditure, using income as a ranking metric might inflate regressivity outcome. Expenditure-based ranking, therefore, is recommended. Additionally, the OECD's study recommended a more in-depth analysis by taking into account the employment status of individuals contributing to household income. Notably, students, the self-employed, and retirees might possess more substantial lifetime earnings and might not be subject to pronounced distributional challenges. In this case, relying solely on an income-based ranking method might skew policymakers' evaluations. Recognizing that no single method can perfectly represent the economic status of all households, it becomes crucial to understand the implications of the chosen approach, especially as more countries are adopting carbon pricing as their primary tool for decarbonization.

Against the background, this paper made the following contributions to the literature. First, to understand the distributional effects of carbon pricing on households, previous review papers had rarely delved into the nuanced distinctions and impact of income and expenditure-based approaches in carbon pricing distributional analysis (Ohlendorf et al., 2021; Wang et al., 2016). Furthermore, the literature addressing this issue was often deficient in empirical data analysis (Köppl and Schratzenstaller, 2023; Pizer and Sexton, 2019). Consequently, this study employed Taiwan's household survey data to explore the disparities and impacts associated with these two methodologies. While the immediate context of Taiwan offered tangible data, the implications of this paper's findings extended beyond its borders. Second, while numerous studies have investigated the distributional effects of carbon pricing utilizing both income and expenditure metrics (Beugin et al., 2016; Mathur and Morris, 2014; Wier et al., 2005), there remains a lack of comprehensive articulation distinguishing between these two approaches in determining carbon burdens. A common theme across these studies was the preference for income as the primary criterion for household ranking, with less attention paid to expenditure as an alternative or complementary metric. This consistent oversight indicated a literature gap concerning the impact of different ranking methods on the assessment of distributional effects. The study by Brys (2014) underscored the significance of exploring various ranking methods, yet empirical analyses leveraging real-world data remain limited. Expanding the analytical horizon, Rausch et al. (2011) offered a multi-faceted investigation of carbon burden in relation to income, race, ethnic group, and region, enriching the discourse on household classification. Although their analysis did not compare income with expenditure ranking methods, it presented a comprehensive approach to evaluating how different household characteristics affect carbon burden assessments. Thus, this research not only clarified the distinctions between carbon burden criteria but also engaged in a robust examination of diverse ranking methods as evidenced in actual data scenarios. Third, in alignment with OECD recommendations, this study integrated an analysis of employment status, a variable previously underexplored in the literature on the distributional effects of carbon pricing (Brys, 2014; Flues and Thomas, 2015). This inclusion provided a more holistic view of the socioeconomic factors in the impacts of carbon pricing, thereby enhancing the depth and relevance of our findings.

Table 2
OECD study recommended ranking method based on households' lifetime economic status and current income/expenditure (Brys, 2014).

Lifetime	Low current income, High current expenditure	High current income, Low current expenditure
Lifetime Rich	Expenditure-ranked	Income-ranked
Lifetime Poor	Income-ranked	Expenditure-ranked

2. Methods and data

2.1. Methodological framework

This study's methods, as depicted in Fig. 1, recognized that carbon pricing impacts household expenditure through two primary channels: direct and indirect emissions. Direct emissions pricing involves charging households based on carbon emissions resulting from their direct activities. This paper accounted for the carbon emissions stemming from household electricity consumption and transport fuel usage as the primary sources of these direct emissions (Section 2.4). Conversely, indirect emissions pricing considers the carbon emissions generated during the production processes of goods and services. By analyzing the carbon intensity across diverse industries using electricity emissions data, this study employed an Input-Output (IO) model paired with data sourced from the Survey of Family Income and Expenditure (SFIE) to assess household indirect emissions (Section 2.5). After setting the carbon prices (Section 2.6), this paper proceeded to quantify household carbon expenditures and analyze their distributional implications (Section 2.7).

For countries lacking a comprehensive dataset regarding carbon intensity at the industry level, assessing the indirect effects of carbon pricing can effectively be done by using the electricity sector as an indicative measure. The methodological framework proposed in this study was designed for adaptability, allowing for its application across different national settings while ensuring rigorous evaluations of distributional outcomes.

2.2. Case study context: carbon pricing policy in Taiwan

This paper contributed to informing the ongoing policy development in Taiwan. Taiwan's economy heavily relies on the export-oriented electronics industry. Historical subsidies for fossil fuels aimed at fostering economic growth have led to high energy consumption and carbon emissions within its industrial sectors, hindering the transition towards a low-carbon economy. Yet, there is increasing pressure on Taiwanese industries and policymakers to adopt carbon schemes to maintain competitiveness in the international markets (Chou and Liou, 2023; Liu and Chao, 2023). On January 10, 2023, Taiwan ratified the "Climate Change Response Act," which set a legally-binding goal of

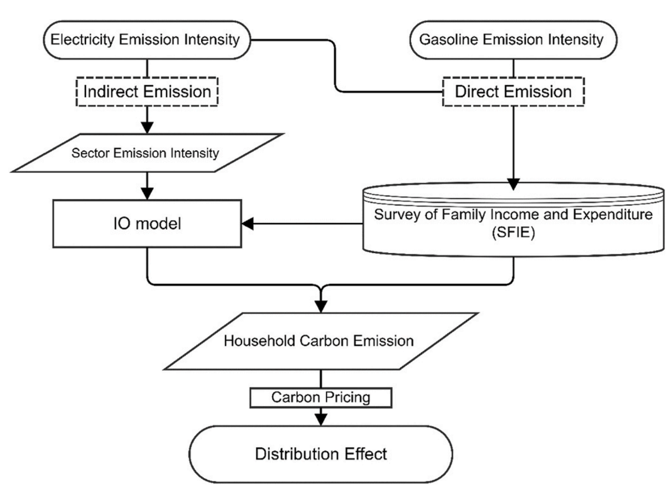

Fig. 1. Flow chart for the proposed methodology.

2050 net-zero and prominently features a proposal on carbon fees (Ministry of Environment, 2023). While the specifics of fee collection and rates are still under development, a 2020 report from the Grantham Research Institute recommended initiating carbon pricing in Taiwan at US$10 per ton, increasing to US$98 per ton by 2030 (Grantham Research Institute on Climate Change and the Environment and Vivid Economics, 2020). Despite Taiwan being at a crucial turning point in its carbon pricing policy, there is a notable absence of in-depth research exploring the potential implications on households. The few existing research studies have indicated a regressive trend in Taiwan's electricity burden, with growing disparities in electricity consumption tied to income (Huang, 2019), rendering low-income households especially susceptible to energy price fluctuations (Huang, 2014). The most relevant investigation on this topic is the research by Feng et al. in which the authors ventured into analyzing the distributional impacts of carbon taxation in Taiwan, emphasizing corporate profits and capital gains; their findings revealed slightly progressive outcomes (Feng et al., 2020). Their analysis, however, did not encompass consumption patterns across varying income brackets, leaving a vital aspect of the analysis untouched. As public sensitivity to energy prices could cast doubts on the long-term viability of the impending policies (Boyce, 2018), this paper aimed to elucidate the distributional effects of carbon pricing schemes to inform policy design.

2.3. Data source and description

This research drew on Taiwan's 2021 Survey of Family Income and Expenditure (SFIE), conducted by the Directorate-General of Budget, Accounting, and Statistics (DGBAS) in 2021 (DGBAS, 2022). The SFIE offered a comprehensive view detailing household demographics, income, expenditure, locations, educational backgrounds, and occupations for more than 15,000 households. While expenditures were originally categorized into 243 distinct groups, for the purpose of analysis, this study consolidated similar categories, resulting in a focused breakdown of 25 expenditure groups. Both income and expenditure data were divided into ten deciles, with each decile comprising an equal number of households. This study also integrated the 2016 Taiwan Input-Output Tables (IOTs) from the DGBAS, which was the most updated version (DGBAS, 2016a). These tables contained relationships between 63 different sectors and can facilitate the creation of an input-output model. The challenges lay in aligning the IO sectors with specific household consumption items, a task addressed using the relationship elucidated in the 2016 Input-Output Statistics report by the DGBAS (DGBAS, 2016b, 2020). This report elucidated which sector in the IOTs the consumption items from the Household Expenditure Survey were categorized under. A nuanced breakdown of the relationship between the IO Table sectors and the SFIE dataset is assessable in the Supplementary Information Table S1.

2.4. Direct emissions assessment

For this study, the assessment of direct emissions sourced its data from the SFIE, specifically focusing on household electricity and gasoline consumption metrics. The emission intensity coefficient for electricity was 0.509 kgCO2e/kWh for 2021, as released by the Bureau of Energy (Bureau of Energy, 2021). Concurrently, the electricity pricing data from the Taipower Company averaged NT$2.6253 (US$0.09) per kWh in 2021 (TaiPower Company, 2021a, b). The carbon intensity metric for gasoline, 2.2631 kgCO2/L, was derived from the Greenhouse Gas Emission Coefficient Management Table published by Taiwan's Environmental Protection Administration (The Environmental Protection Administration, 2019).

The SFIE dataset bundled gasoline consumption and parking fees while making no distinction between car and scooter use. Given the evident differences in parking fees and fuel (gasoline) expenses between these vehicle types, this study referenced the 2021 Small Private Passenger Cars Usage Survey and the 2021 Motorcycle Usage Survey by the Ministry of Transportation and Communications (MOTC) to discern the ratio of fuel expenses to parking fees (Ministry of Transportation and Communications, 2021a, b). In addition, with the aid of household vehicle data from the SFIE, this study refined the gasoline consumption figure, offering a more accurate depiction of household gasoline expenses.

2.5. Indirect emissions: the input-output (IO) model approach

To estimate the additional expenditures due to indirect emissions, this study applied the input-output (IO) model, a tool that captures the interconnections among industries (Leontief, 1986). This method has the key assumption of a linear relationship in the carbon payments made by each sector. It posits that carbon prices do not influence the input quantities of raw materials, meaning the carbon payments incurred during production are entirely passed on to the final product prices.

This study integrated Taiwan's Input-Output Tables (IOTs), which encompassed 63 industries, with the Survey of Family Income and Expenditure (SFIE) data to determine household expenditures across sectors. By leveraging the environmental extended IO model through the Leontief equation, a widely-used method in multiple related studies (Grainger and Kolstad, 2010; Jiang and Shao, 2014; Wier et al., 2005; Yan and Yang, 2021), this paper derived household indirect carbon emissions ($E^{indirect}$) as follows:

$$E_f^{indirect} = TK(I-A)^{-1}F_f \tag{1}$$

- $E_f^{indirect}$ represents an 1×25 vector, signifying the indirect carbon emissions generated by household f across various expenditure categories;
- T denotes the carbon price;
- K corresponds to an 1×63 vector, denotes sectoral carbon intensity, signifying carbon emissions per dollar spent;
- $(I-A)^{-1}$ stands for the Leontief inverse matrix, a 63×63 matrix illustrating industry interrelations and quantifying output essential to fulfill a unit of final demand;
- F_f is a 63×25 matrix, illustrating household f 's demand for the 25 expenditure categories allocated by sector;

The Grantham Research Institute on Climate Change and the Environment, in its 2020 publication, "Carbon Pricing Options for Taiwan," advocated for encompassing carbon emissions from the entire electricity generation sector in Taiwan's carbon pricing to accurately capture indirect emissions spanning the residential, commercial, and service domains (Grantham Research Institute on Climate Change and the Environment and Vivid Economics, 2020). The literature emphasized the pivotal role of the electricity sector in accumulating and distributing carbon charges (Dorband et al., 2019; Steckel et al., 2021). Owing to the unavailability of carbon intensity data for various industries in Taiwan, this study narrowed its focus on the energy consumption of each industry when assessing carbon fee additions for indirect emissions. Here, the electricity sector was considered the primary point for fee accrual, allowing for the estimation of carbon intensity coefficients across industries. As such, the carbon intensity (K) is defined by the following expression:

$$K = \frac{K_e}{P_e} A_e \tag{2}$$

- K corresponds to aa 1×63 vector, denotes sectoral carbon intensity, signifying carbon emissions per dollar spent (tCO2/NT$);
- K_e denotes the carbon intensity specific to the electricity industry (tCO2/kWh);
- P_e is the electricity price (NT$/kWh);

- A_e is a 1×63 vector, denotes the proportion of electricity expenses in various sectors, derived from the Leontief matrix within the IOTs;

2.6. Carbon price determination

This study adopted a carbon price of US\$40/tCO_2 as the foundational scenario, which aligns with the 2020 baseline recommendation by the Carbon Pricing Leadership Coalition (CPLC) to realize the Paris Agreement goals by 2050 (Stiglitz et al., 2017). Complementing this, this paper also conducted sensitivity assessments with two alternate carbon prices. First, This study used US\$10/tCO2 based on the initial recommendation in the "Carbon Pricing Options for Taiwan" report (Grantham Research Institute on Climate Change and the Environment and Vivid Economics, 2020). Second, a higher carbon price of US\$100/tCO2, as suggested by the CPLC for realizing the Paris Agreement targets (Stiglitz et al., 2017). This study used the currency exchange rate of NT \$27.932 per US dollar, as the United States Internal Revenue Service reported for the average currency exchange rate in 2021 (IRS, 2023).

2.7. Distributional effects of carbon pricing

This study analyzed the distributional effects of carbon pricing based on various assessment criteria. For household categorization, two distinct methods were applied: ranking by income and by expenditure. In evaluating the regressivity of carbon pricing, this paper considered two foundational baselines: the income-based approach and the expenditure-based approach. As a result, four distinct evaluation criteria emerge, namely.

1. Income ranking with income as a base (Income-ranked, Income Base)
2. Income ranking with expenditure as a base (Income-ranked, Expenditure Base)
3. Expenditure ranking with income as a base (Expenditure-ranked, Income Base)
4. Expenditure ranking with expenditure as a base (Expenditure-ranked, Expenditure Base).

2.7.1. Income versus expenditure: a critical distinction

When discussing distributional effects, it is essential to highlight two determinants: First, the criteria for distinguishing between low-income and high-income groups, and second, the foundation for assessing the weight of the tax burden. As the OECD research suggested, there are two primary choices for these determinants: income and expenditure (Brys, 2014; Flues and Thomas, 2015). This results in four assessment combinations.

1. Evaluate tax burden as a percentage of income, ranked by income.
2. Evaluate tax burden as a percentage of income, ranked by expenditure.
3. Evaluate tax burden as a percentage of expenditure, ranked by income.
4. Evaluate tax burden as a percentage of expenditure, ranked by expenditure.

Fig. 2 shows the ratio of expenditure to income. For households ranked by income, the lowest quantile (representing the poorest) spends more than they earn, implying negative savings. This ranking also revealed an inverse relationship between income level and expenditure ratio: higher-income households tend to spend a smaller proportion of their earnings. In contrast, ranking by expenditure resulted in more consistent expenditure ratio patterns. Different household income levels presented distinct consumption characteristics, underscoring the importance of considering both ranking approaches in distributional effect analyses.

2.7.2. Evaluating carbon pricing progressivity: The Suits Index

To analyze the distributional implications of carbon pricing—-whether it is regressive or progressive—this study employed the Suits Index. This index, proposed by economist Daniel B. Suits in 1977, assesses tax regressivity across income groups (Suits, 1977). A significant body of research on the distributional effects of carbon pricing has utilized the Suits Index for its assessment (Berry, 2019; Hardadi et al., 2021; Wang et al., 2019; Yan and Yang, 2021).

Fig. 3 portrays the Suits Index. While the y-axis represents the cumulative percentage of carbon expenditure, the x-axis shows the cumulative percentage of income from low to high. The line OB indicates an even distribution of carbon expenditure across income groups.

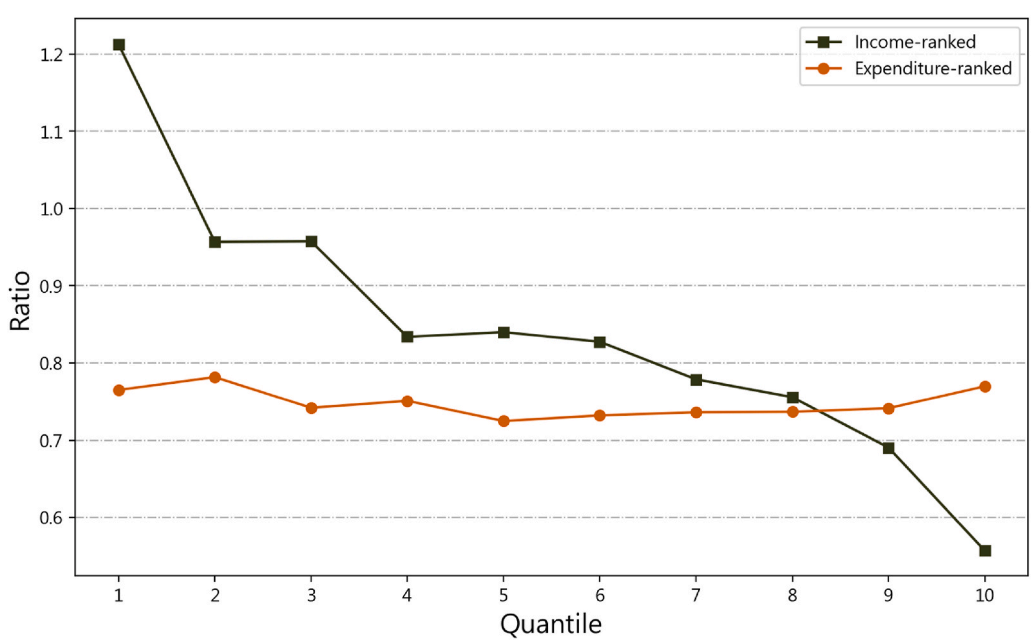

Fig. 2. Ratio of household expenditure to income across various income level groups (data taken from (DGBAS, 2022)).

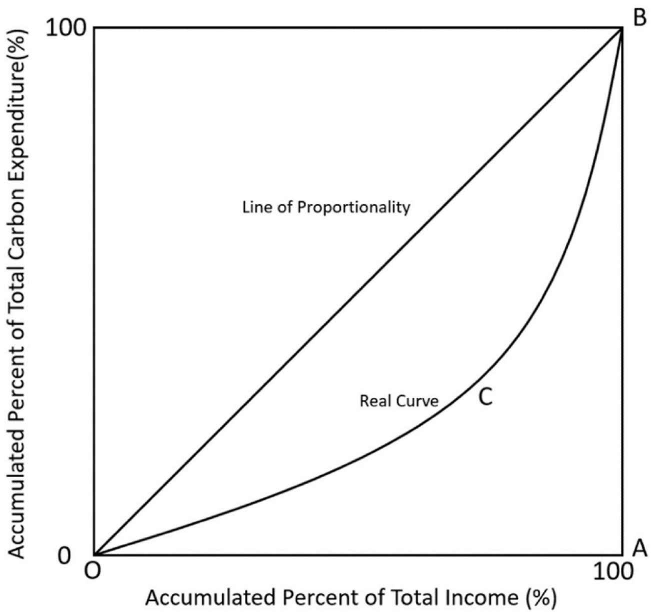

Fig. 3. Illustration of the suits index.

Meanwhile, the curve C represents the real expenditure distribution. The Suits Index is computed as follows:

$$S\ (Suit\ Index) = \frac{(K-L)}{K} = 1 - \frac{L}{K} \tag{3}$$

- L denotes the area below Curve C;
- K refers to the area of OAB, consistently at 0.5;

The Suits Index (S) value lies between −1 and 1. For an extremely regressive tax system, where all tax revenues are from the low-income bracket, the area L becomes 1, and the S value is −1. Conversely, in a highly progressive tax setup, where the tax revenues solely come from high-income earners, the area L is equal to 0, making S equal to 1. If tax revenue is evenly distributed across all income groups, L matches K, making S equal to 0. Therefore, an S value less than 0 indicates a regressive tax, while an S value greater than 0 denotes a progressive one. Since K is equal to 0.5, Eq. (4) can be further expressed as follows:

$$S\ (Suit\ Index) = 1 - \sum\nolimits_{h=1}^{n} (T(y_h) + T(y_{h-1})) \times (y_h - y_{h-1}) \tag{4}$$

- h represents the sequential numbering of households based on ascending income, from lowest to highest.
- n is the total number of households;
- y_h represents the cumulative percentage of income with the sequence number h;
- $T(y_h)$ is the cumulative carbon expenditure when the cumulative percentage of income is y_h;

Given the assessment combinations described in Section 2.7.1, the definition of y_h varies. Depending on whether income or expenditure ranking is chosen, h represents the sequential number of households based on the chosen ranking method. With income or expenditure as the base, y_h denotes the cumulative percentage of the selected base with the sequence number h.

3. Result and discussion

In the subsequent subsections (3.1–3.3), the analysis primarily focused on the distributional effects arising from a US$40 carbon price. To provide a more comprehensive perspective, Section 3.4 extended this examination by contrasting the impacts across three distinct carbon pricing scenarios: US$10, US$40, and US$100.

3.1. Distributional results using income-ranked and income base

Before conducting a comparative analysis using various evaluation criteria, this study first applied the Income-ranked and Income Base—benchmarks commonly used in previous studies—to understand the distributional implications of carbon pricing in Taiwan. This approach provided not only a point of comparison with existing literature but also offered a preliminary insight into the interplay between carbon consumption tendencies and the economic brackets of Taiwanese households.

Fig. 4 illustrates that the impacts of carbon pricing on Taiwanese households vary from approximately NT$5077 (US$182) in Quantile 1 to NT$21,927 (US$785) in Quantile 10, with an average of NT$12,370 (US$443). A superficial look focusing on absolute carbon expenditures would suggest that the wealthier groups shoulder a more substantial carbon burden. However, a more holistic view requires considering carbon expenditures in relation to household income. When viewed as a proportion of income, the impacts range from around 1.9% at Quantile 1 to 0.8% at Quantile 10, averaging out at 1.3%. This revealed a pronounced regressive characteristic of carbon pricing, with lower-income households bearing a heavier relative burden. Specifically, the financial impact of carbon pricing on the lowest quantile is over twice as significant as that on the highest. Previous studies reported similar distributional effects ranging from 0.5% to 3% across different countries (Dorband et al., 2019; Steckel et al., 2021; Wang et al., 2016), reinforcing the validity of the observations made in this study.

Fig. 5 provides a breakdown of direct versus indirect emissions expenditures. As income increases, there is a notable shift towards a higher proportion of indirect emissions expenses. For Quantile 1, indirect emissions account for about 60% of expenses, a figure that climbs to 70% for Quantile 10. This trend is anticipated, given that indirect emissions encompass a broad spectrum of non-essential expenditures, like recreation and services, while direct emissions come largely from transportation fuel and electricity bills. However, a detailed understanding requires a more granular look at spending categories.

Fig. 6 displays the top five carbon expenditures across households, while all other expense items were collectively categorized as 'Other items.' A notable pattern emerged: food expenses, as a fraction of total expenses, decreased with increasing income. This observation aligned with Engel's law, suggesting that the proportion of expenditure dedicated to food increases as income drops. Notably, Engel's Law is not confined to food; it also appears to extend to electricity, a fundamental necessity. Given the high carbon footprint of electricity generation, such a trend in carbon expenses, tied to income, necessitated policymaker attention.

Conversely, gasoline expenses exhibited the opposite trend. Households with higher incomes dedicated a more significant portion of their expenditures to gasoline, ranging from 9% for Quantile 1 to about 13% for Quantile 10. This pattern could be attributed to a greater propensity among wealthier individuals to own private vehicles, while those with lower incomes often rely more on public transportation and two-wheelers. This observation aligned with prior studies consistently demonstrating a positive correlation between income levels and gasoline expenditure percentages (Berry, 2019; Ohlendorf et al., 2021; Steckel et al., 2021; Yan and Yang, 2021).

In addition, the proportion of expenditures under the 'Other items' category grew with increasing income. This indicated that affluent households had a broader and more varied spending pattern. Recognizing this variability in indirect emissions related to spending was vital to designing carbon pricing policies that were both equitable and just.

The tendency for carbon pricing to display regressive effects in higher-income countries has been anticipated in the literature (Ohlendorf et al., 2021; Steckel et al., 2021; Wang et al., 2016). This study's

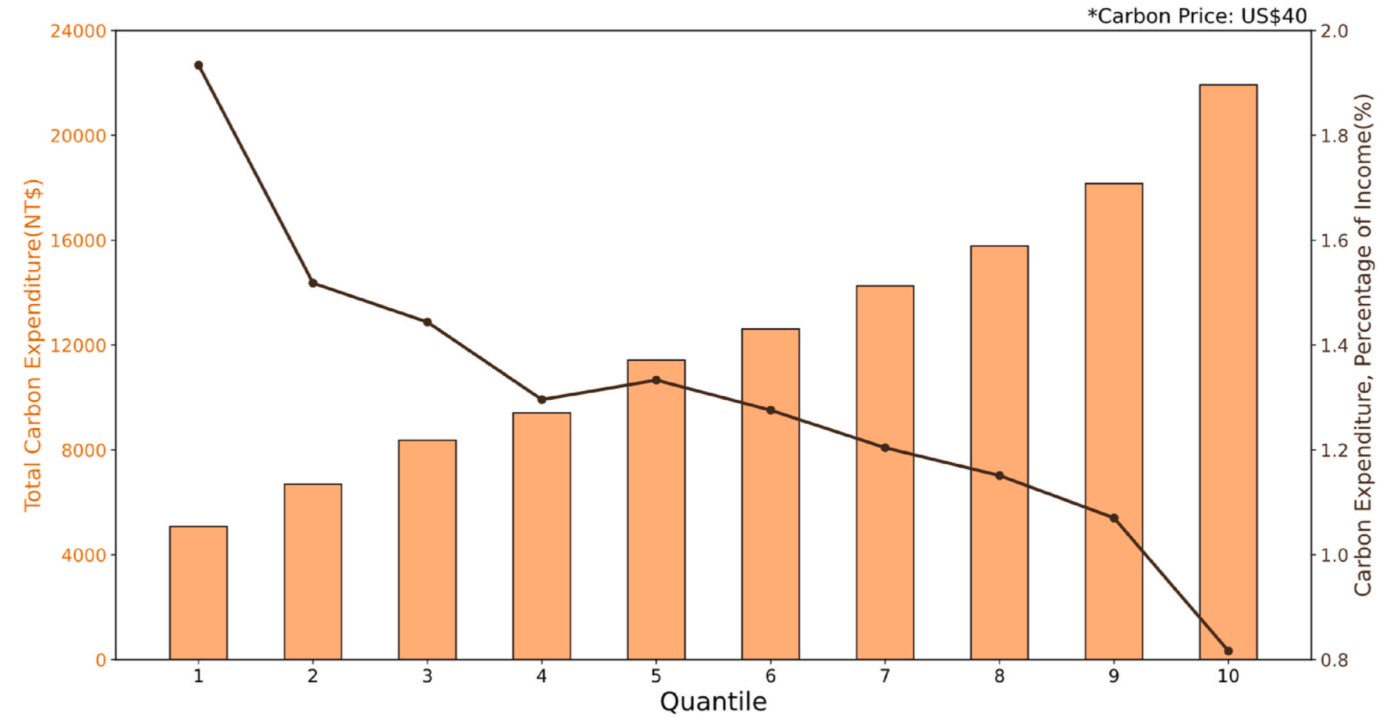

Fig. 4. Carbon expenditure and relative burden (as % of income).

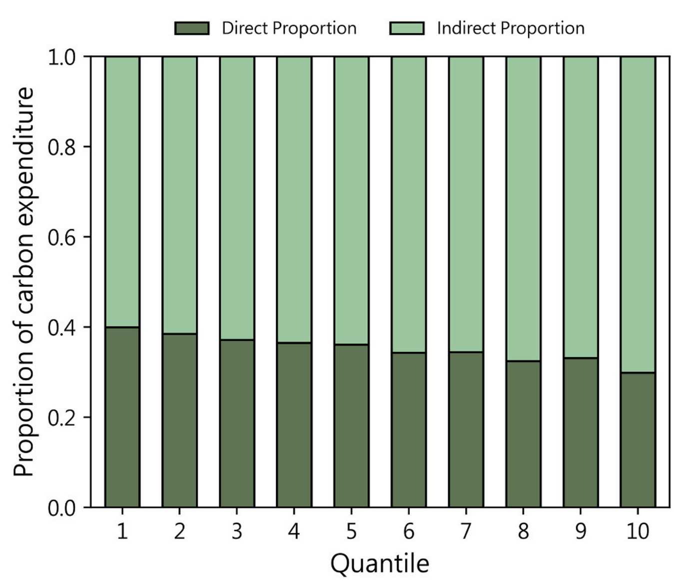

Fig. 5. Breakdown of direct versus indirect carbon expenditures.

focus on Taiwan corroborated such a regressive pattern, in line with findings from similar economies. Notably, Grainger and Kolstad (2010) study in the United States found the carbon burden for low-income individuals to be 2–3 times that of their high-income counterparts. Our findings echoed this, showing that in Taiwan, low-income individuals bear a carbon burden about twice that of high-income individuals. Prior studies have linked this regressive pattern to energy consumption (Dorband et al., 2019), examining the effects on household electricity and transportation fuel (Berry, 2019; Verde and Tol, 2009). Our analysis, with its comprehensive consumption data, indicated that while electricity and transportation are indeed significant, their impact is relatively limited within high-income households—who face a more varied carbon burden across numerous categories of expenditure.

3.2. *Evaluating the distributional effects of different criteria*

As evidenced in Section 3.1, the distributional effects of carbon pricing in Taiwan leaned towards a pronounced regressive trend. However, as underscored in the literature review, the choice of evaluation criteria - specifically, different combinations of ranking and base - could significantly influence these outcomes.

Fig. 7(a) shows the distributional effects of carbon pricing using both the income-based and expenditure-based assessment approaches when ranked by income. The pronounced regressive pattern observed using the income-based method became less distinct when the metric shifted to an expenditure-based method. The average results derived from the expenditure-based method were notably higher, standing at 1.55%, compared to the 1.3% obtained using the income-based method. Fig. 7 (b), which ranks households based on expenditure, depicted a marginally regressive pattern for both income-based and expenditure-based assessments. The average distributional effect when using expenditure as both rank and base was 1.56%, closely mirroring the results of the income rank with expenditure base. However, when employing the expenditure rank with an income base, the average result was slightly reduced compared to that derived from the income rank approach. Evidently, the expenditure-based evaluation method offers more consistent outcomes, irrespective of the ranking approach, thereby exhibiting resilience against methodological variations. For a more comprehensive understanding of these regressive trends, it becomes essential to compute the Suits Index, enhancing the comparability of results.

Table 3 summarizes the Suits index calculations for various carbon expenditures, encompassing direct (specifically from electricity and gasoline consumption emissions) and indirect emissions, guided by the four evaluation methods detailed in Section 2.6. Several primary observations emerged from these findings.

1. *Regressive Trends*: The “Income-ranked, Income Base” method distinctly exhibited regressive tendencies, as evidenced by its Suits index value of −0.1207. In contrast, the “Income-ranked, Expenditure Base” approach demonstrated minimal regressivity, with its

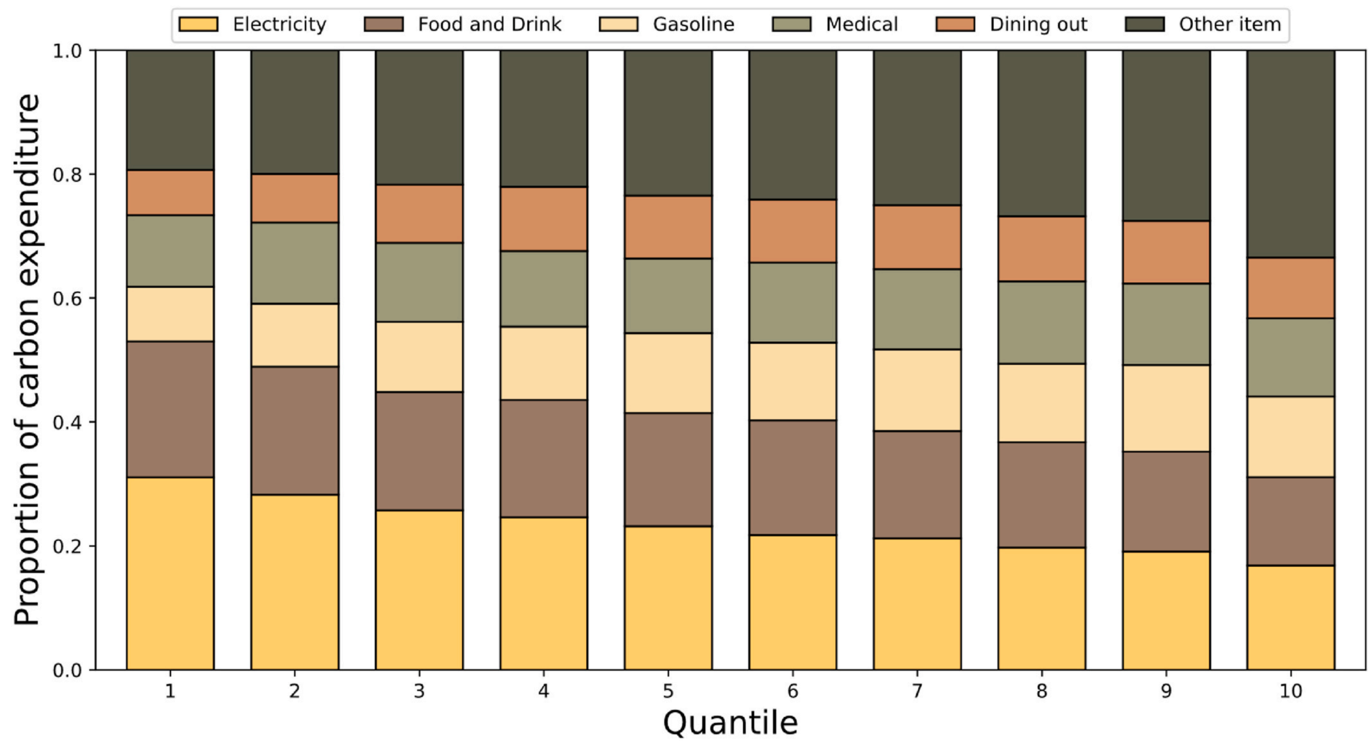

Fig. 6. Breakdown of top five carbon expenditures across households with remaining expenses grouped as Other Items.

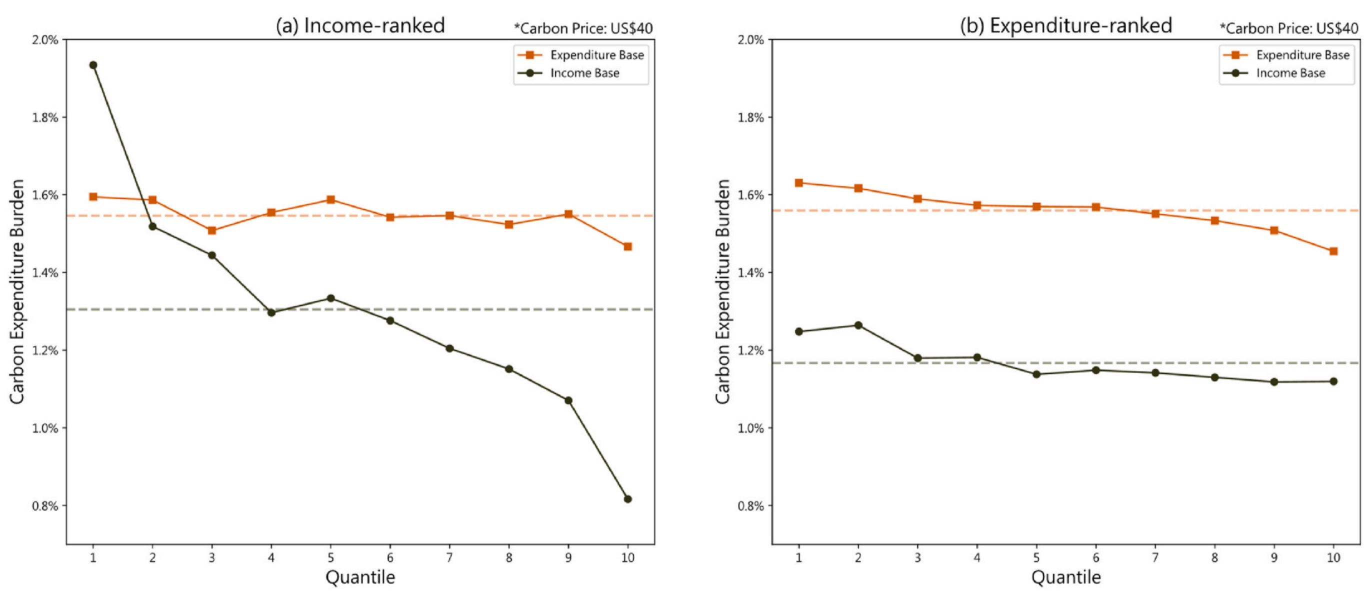

Fig. 7. Carbon burden rate when households are ranked by (a) income, (b) expenditure.

Suits index resting at just −0.0109. This distinction is supported visually in Fig. 7.

2. *Alignment with Existing Literature*: Focusing on the often-cited “Income-ranked, Income Base” framework, observations of this paper are consistent with established literature (Feng et al., 2010; Kerkhof et al., 2008; Steckel et al., 2021; Verde and Tol, 2009). A pronounced regressive trend is evident, especially when assessing direct emissions and carbon expenditures linked to electricity. Meanwhile, the regressivity observed for indirect and gasoline-associated emissions is quite small.
3. *Variability Across Criteria*: Transitioning from the pronounced regressivity of the “Income-ranked, Income Base” method, the other criteria portrayed a significant softening of this trend, with Suits index values mainly oscillating between −0.01 and −0.02. The analysis revealed notable shifts in the Suits Index, suggesting significant changes in the distributional effects of carbon pricing when different bases for carbon burden are applied. For detailed statistical results, including confidence intervals and *t*-test comparisons, readers are directed to Table S2 in the Supplementary Information. Such disparate outcomes highlighted the pivotal role played by the chosen assessment criterion.
4. *Direct vs. Indirect Emissions*: When evaluating direct emissions, outside of the “Income-ranked, Income Base” paradigm, the other methods showcased milder regressivity, all with Suits index values

Table 3
Suits index analysis for carbon price payment, 2021 (Note: Orange shading indicates Suits Index values below −0.1[a.], while green shading denotes values greater than 0).

	Income-ranked, Income Base	Income-ranked, Expenditure Base	Expenditure-ranked, Income Base	Expenditure-ranked, Expenditure Base
Direct	-0.1654	-0.0566	-0.0905	-0.0949
- Electricity	-0.2124	-0.106	-0.1365	-0.1403
- Gasoline	-0.0847	0.0282	-0.0116	-0.0169
Indirect	-0.0976	0.0128	0.0252	0.0196
Total	-0.1207	-0.0109	-0.0142	-0.0194

[a.] The benchmark of -0.1 has been historically significant in the literature, serving as a demarcation point to signify pronounced regressivity. Previous studies had consistently discussed Suits Index values below -0.1 as indicative of significant regressive outcomes (Andersson and Atkinson, 2020; Wang et al., 2019). This threshold has been a commonly observed data for substantial regressivity across various assessments of taxation impacts (Berry, 2019; Hardadi et al., 2021; Wier et al., 2005).

falling below −0.1. Conversely, indirect emissions presented a progressive trend in all but the "Income-ranked, Income Base" criterion, indicated by positive Suits index values.

5. *Electricity vs. Gasoline Expenditures*: All methodologies led to a regressive pattern for electricity-related carbon emissions, underscoring its role as a dominant carbon emitter and essential daily requirement. Gasoline-associated expenditures, however, presented a mixed picture. While the "Income-ranked, Expenditure Base" method hinted at a progressive trend (Suits index at 0.0282), the other criteria show only mild regressivity, with values staying below −0.1. This aligns with existing research (Andersson and Atkinson, 2020; Wier et al., 2005; Yan and Yang, 2021) that depicted a fluctuating pattern for gasoline, ranging from regressive to progressive tendencies.

Previous studies that employed both income and expenditure as bases for assessing carbon burden have provided valuable insights into the regressivity of carbon pricing. For instance, Wier et al. (2005) analyzed the regressive nature of carbon burden by comparing these two bases. Despite uncertainties in their expenditure data, their study indicated a trend towards reduced regressivity when the expenditure base was used. This observation was similarly noted in the works of Mathur and Morris (2014), and Beugin et al. (2016). These studies consistently demonstrated that an expenditure-based assessment tends to yield a more proportional and, at times, progressive impact of carbon burden compared to an income-based assessment. However, a gap in these studies was the lack of a concrete metric to measure regressivity. Our study addressed this gap by employing the Suits Index to quantify the degree of regressivity across different carbon burden bases. Furthermore, a unique aspect of our research lies in the exploration of various household ranking methods. Prior literature predominantly focused on income-based household ranking, but our study extended this exploration by integrating and analyzing different ranking methodologies. This approach not only complemented existing research but also enhanced our understanding of how different ranking methods can influence the perceived distributional effects of carbon pricing.

From the findings of this study, it's evident that the chosen assessment base greatly determines the observed regressivity or progressivity in carbon expenditure. When using the widely accepted "Income-ranked, Income Base" as a reference, the other three methodologies not only tempered the observed regressivity but can even render progressive results. Most notably, the "Income-ranked, Expenditure Base" method yielded the most substantial decrease in regressivity. Past research ascribed these discrepancies to variances in employment status and consequent consumption behaviors across income brackets (Brys, 2014; Flues and Thomas, 2015).

3.3. *Impact of employment status*

Informed by insights from the previous section, it becomes imperative to delve deeper into how employment status nuances influence carbon expenditure patterns within household financial dynamics. Fig. 8 illustrates the differing narratives presented when households were classified by income versus expenditure. Fig. 8(a), which categorizes households by income, displays a pronounced, near-exponential relationship between income and savings. In contrast, Fig. 8(b) showcases a more balanced growth when ranked by expenditure. Such consistent expenditure trends, regardless of the ranking method, suggested possible variances in carbon burden interpretations. Leveraging OECD insights (Brys, 2014) on the importance of lifetime wealth in selecting an equitable ranking methodology, employment status stood out as a critical determinant. Thus, this paper pivoted the analysis towards understanding carbon expenditure in the context of diverse employment scenarios.

3.3.1. *Relationship between employment status and ranking criteria*

Pizer and Sexton (2019) suggested that household expenditure is a more accurate reflection of long-term economic status, advocating for its use in household ranking. The OECD study offered a more complex view, asserting that both immediate and long-term economic conditions should be considered in the ranking process (Brys, 2014). Our study expanded on these discussions by incorporating actual employment data to examine how different ranking methods influence household economic mobility. This approach allowed us to ground theoretical debates in empirical data, highlighting the real-world implications of ranking choices on distributional outcomes.

When various ranking criteria were employed, certain households, specifically those with contrasting income and expenditure profiles,

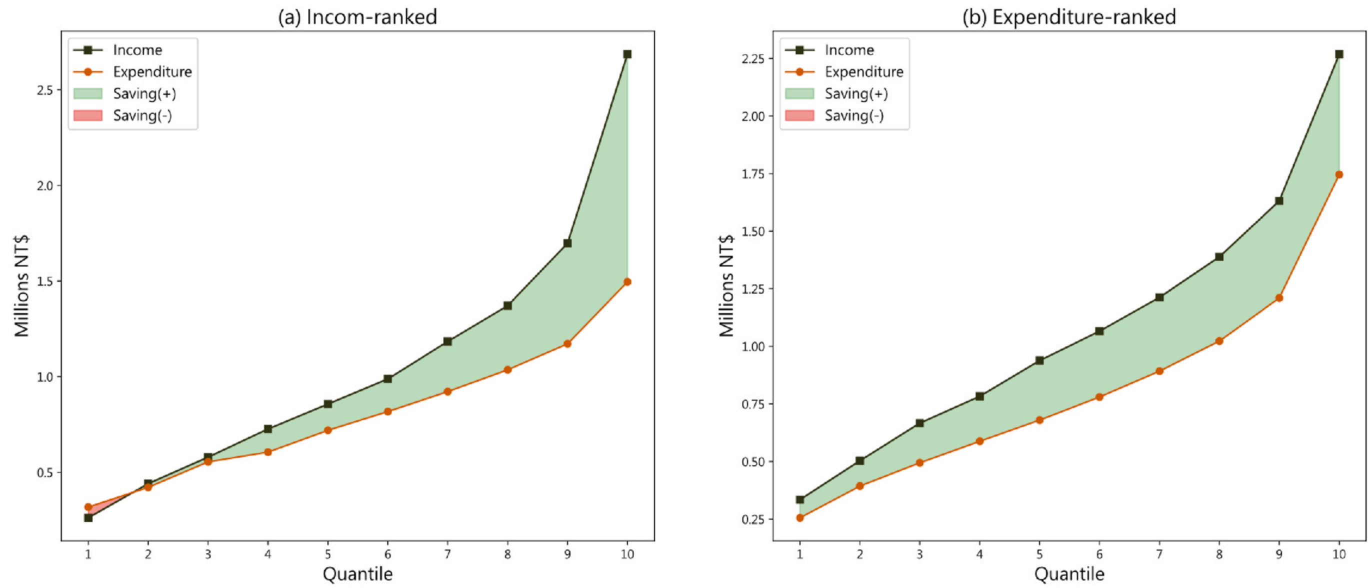

Fig. 8. Financial profiles of households: (a) Ranked by income; (b) Ranked by expenditure.

stood out. These households might fall into different economic categories based on the chosen ranking method, which in turn could alter the outcomes of distributional analyses. Here, employment status acted as a crucial yardstick to analyze shifts across different ranking methodologies. Notably, categories like non-working professionals, self-employed individuals, and retirees might display modest current incomes, yet they could have considerably higher accumulated earnings over their lifetime. Therefore, ranking these households based solely on their current income might lead to a skewed perception of their actual economic standing.

Delving into the SFIE dataset, it classified household employment status into six distinct categories: Employers, Employees, Self-employed, Other, Stay-at-home parents, and Unemployed. Interestingly, the "Other" group encompassed retirees and individuals not actively seeking employment. Such a group is characterized by seemingly low incomes but possibly higher expenditures. Fig. 9 presents ridge plots depicting the distribution of these employment statuses across income quantiles. An expenditure-based ranking saw certain groups, specifically "Other," " Stay-at-home parents," and "Unemployed," shifting to higher quantiles—indicative of these groups typically have lower incomes. Conversely, certain households in the higher-income "Employers" group experience a leftward shifted from higher to lower quantiles. These migration patterns in the data were crucial, as they informed the potential discrepancies observed in distributional outcomes when switching between ranking methods.

SFIE dataset was largely characterized by three categories:

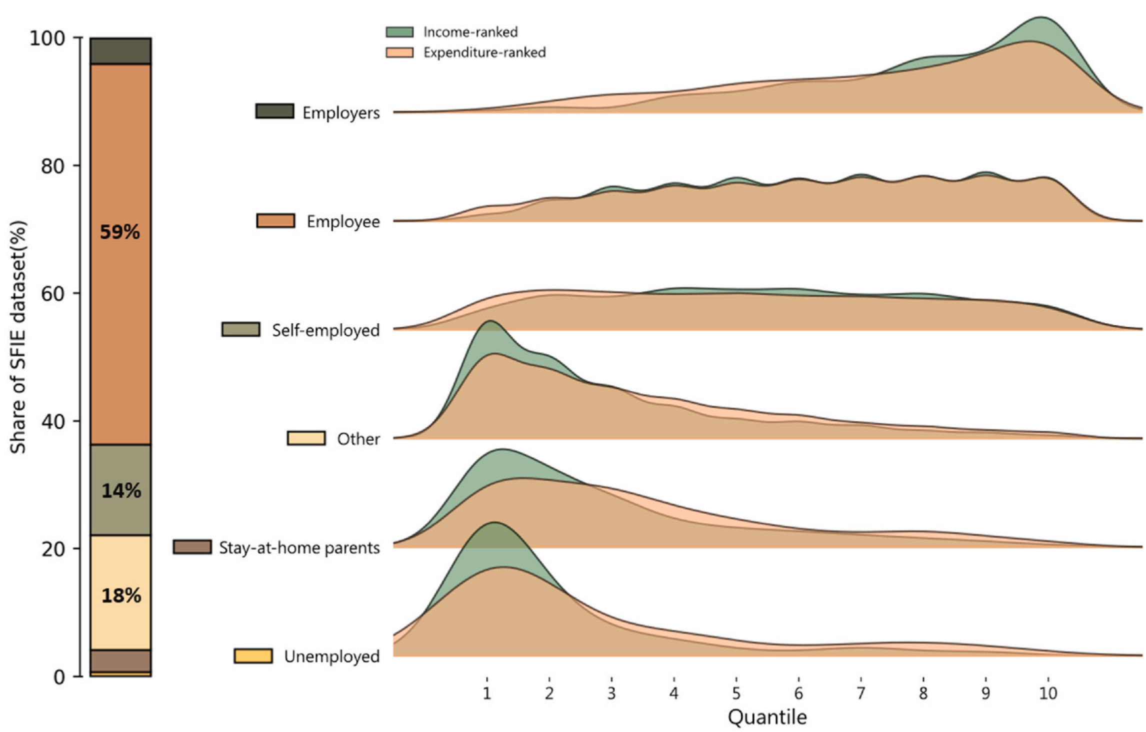

Fig. 9. Distribution of different employment categories across income quantiles.

"Employees," "Self-employed," and "Other," which together account for 91% of the total. Within these, both the "Self-employed" and "Employees" categories consistently showcased similar income and expenditure patterns, frequently landing in identical quantiles across different ranking methodologies. In contrast, the "Other" employment status, which represented 18% of the dataset, significantly drove the observed discrepancies in ranking outcomes. A deeper examination of this category was pivotal to better understanding the overall distributional impacts. Households in the "Other" group tend to mirror the financial characteristics of those with high lifetime wealth, but current low incomes and high expenditures. Such characteristics underlined the importance of employing expenditure-based ranking for a more balanced assessment.

3.3.2. Distributional effects of employment status

With a foundation established on how employment status interacts with ranking methods, this paper subsequently analyzed the distributional effects of carbon pricing with a focus on employment categories. Fig. 10 presents how carbon burdens spread across various employment statuses. When income was utilized as an assessment base, categories like the unemployed, stay-at-home parents, and the "Other" group witnessed substantial variability in impacts. On the other hand, an expenditure-based analysis harmonized the effects across employment categories, leading to more standardized outcomes. Such findings suggested that while categories like self-employed individuals, employees, and employers have synchronized income and expenditure profiles, groups like the unemployed, stay-at-home parents, and within the "Other" category display broader income spectrums but comparable expenditure patterns. Specifically, certain households within the unemployed category may possess relatively higher incomes compared to their unemployed peers but display conservative expenditure patterns, possibly in anticipation of future income uncertainties.

Relying solely on an income-based analysis can distort the perceived economic challenges faced by groups like stay-at-home parents and the "Other" category. On the flip side, low-income households with relatively higher expenditures also emerged, reinforcing the importance of using expenditure as an assessment parameter. Adopting expenditure as the primary metric offered a holistic view, considering factors like wealth and potential borrowing, thereby providing a more comprehensive reflection of a household's long-term economic status.

Fig. 11 provides a breakdown of carbon consumption patterns across different employment categories. With increasing income (from left to right in the figure), there's a noticeable decline in the proportion of electricity expenditure, consistent with Engel's Law. Interestingly, the expenditure behavior of the unemployed appeared to mimic that of their higher-income counterparts, especially in the "Other item" category. Although the unemployed's spending in this category exceeded that of other employment statuses in comparable income brackets, it did not

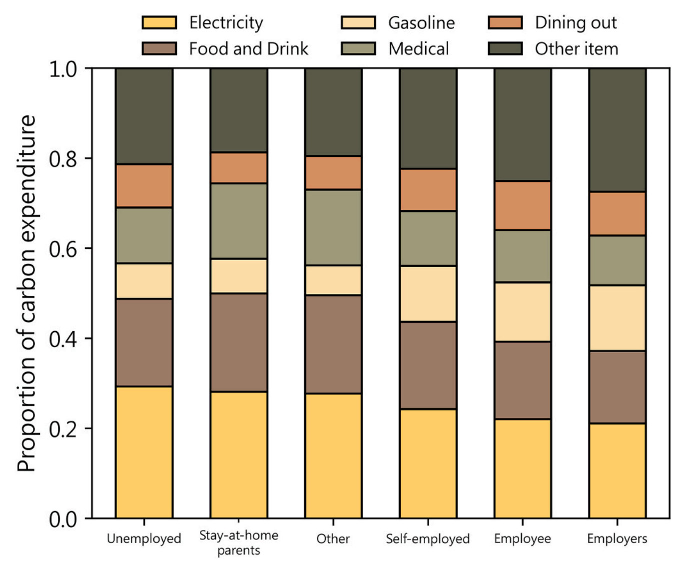

Fig. 11. Carbon consumption patterns across employment categories, segmented by the top five predominant items and aggregated others.

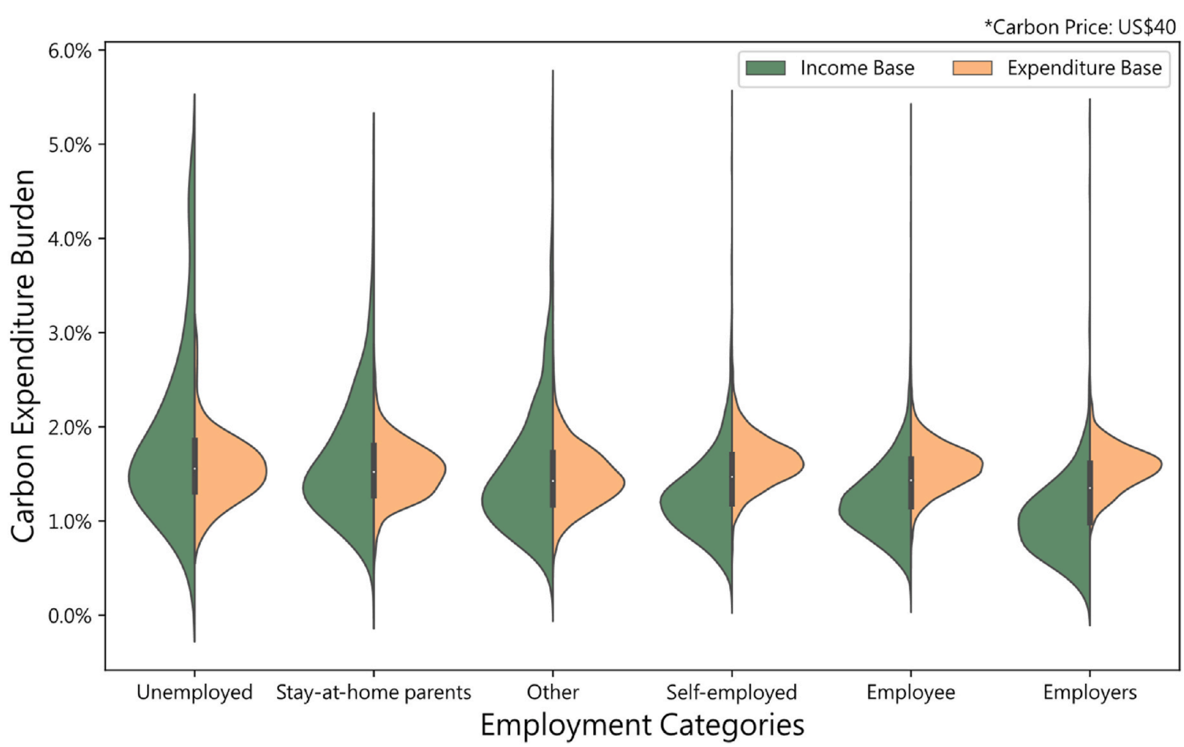

Fig. 10. Distributional effects of carbon burdens across employment categories in income-based and expenditure-based assessments.

mirror the precise consumption habits of the affluent. Data suggested that within the "Other item" segment, the unemployed tended to spend more on items such as tobacco, alcohol, and betel nut, as well as water utilities, than those in the high-income peers.

3.4. Sensitivity analysis of carbon price

While the IO model employed in this study did not factor in the elasticity of carbon emissions in relation to carbon prices, this study conducted a sensitivity analysis to evaluate the distributional effects across a range of carbon price benchmarks: \$10, \$40, and \$100 per ton of CO_2. Fig. 12 presents these effects across diverse income groups for each of these benchmarks, contrasting results from both income-based and expenditure-based assessments.

In the income-based evaluation, carbon expenditures stood at 0.33%, 1.31%, and 3.28% of total income for the \$10, \$40, and \$100 price benchmarks, respectively. In comparison, when evaluated through the expenditure base, the carbon costs constituted 0.4%, 1.55%, and 3.89% of income at the respective price levels. Notably, the income-based analysis consistently showed a regressive trend across all price settings. As the carbon price increased, the lowest income quintile (Quantile 1) shouldered a more significant burden relative to the highest income quintile (Quantile 10). Conversely, the expenditure-based analysis lacked an evident regressive pattern across any of the carbon price levels. Of importance is the observation that as carbon prices rise, the disparity between the findings of the two analyses becomes more pronounced. This highlighted that the decision on which base to adopt for appraising distributional impacts will gain significant relevance as carbon pricing intensifies in the coming years.

4. Conclusions

At the heart of this research lay the debate between income and expenditure as foundational metrics for assessing the distributional effects of carbon pricing. Utilizing a detailed dataset from Taiwan, a nation at a pivotal juncture in its carbon pricing policy decision-making, this study sought to decode the implications of both evaluation bases. This study underscored the pronounced regressive tendencies in Taiwan's carbon pricing when evaluated via traditional metrics—income ranking with income as a base (Income-ranked, Income Base), reflecting a Suit Index of −0.1207. The observed carbon burden on the lowest-income bracket was significantly higher than that on the highest. This trend mirrors prior research, which often found advanced economies showcasing regressive effects in carbon pricing evaluations.

However, the shift to an alternative evaluation base markedly alters the observed distributional effects. The study's findings underscored that the choice of evaluation base — whether for ranking households or assessing carbon burdens — significantly influences the Suit Index, which reflects the regressivity of carbon pricing. Notably, when utilizing expenditure as the base, the Suit Index moderated to approximately −0.01 to −0.02, indicating a less regressive outcome. This approach yielded a consistent portrayal of carbon burdens and their associated regressivity, irrespective of the sorting criterion employed. According to the Permanent Income Hypothesis, households are posited to adjust their current expenditures in anticipation of future economic conditions (Friedman, 1957). Thus, expenditure is argued to be a potentially more reliable measure of a household's long-term financial situation, serving as a valuable evaluative base for carbon burdens (Pizer and Sexton, 2019). The purported stability and representativeness of the expenditure base underscore its importance in analyzing distributional effects. These findings highlighted the potential importance of the expenditure base in directing future research on the distributional impacts of carbon pricing,

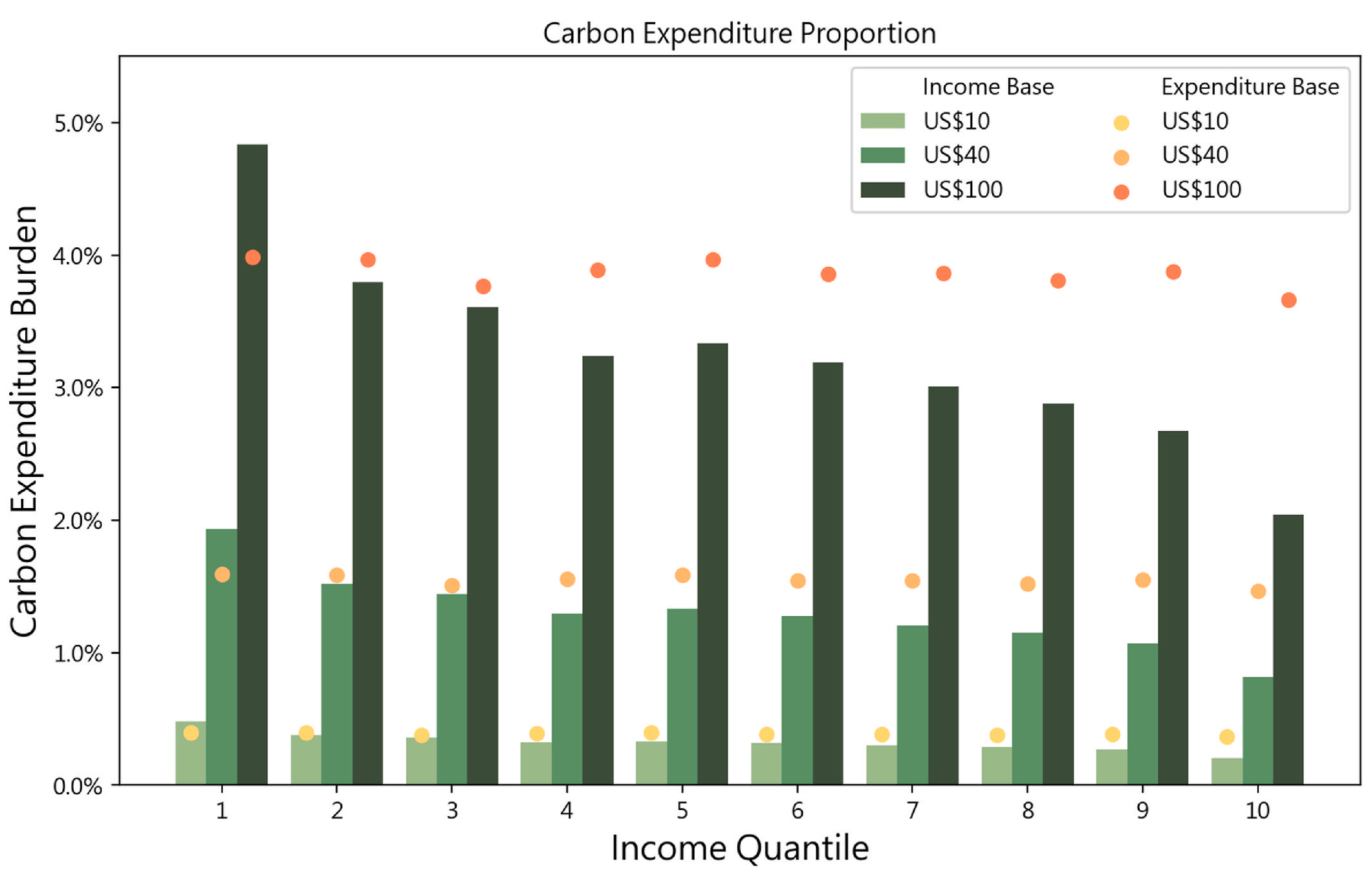

Fig. 12. Comparative distributional effects across income brackets for different carbon price benchmarks using income-based and expenditure-based assessments.

suggesting that it may provide a more equitable assessment across different income groups.

Diving deeper into consumption patterns, this paper noted that electricity consumption, regardless of the assessment lens (income or expenditure), consistently manifested regressive tendencies. This reflected the dominant regressive nature inherent in household energy expenditures, an echo of past studies that emphasize the substantial influence of the energy sector on distributional effects. This alignment with Engel's Law in electricity consumption propelled discussions on the interplay between energy poverty and carbon pricing. Moreover, indirect emissions and gasoline consumption hinted at progressive tendencies, suggesting the inclusion of these components in future carbon pricing designs to ensure a more comprehensive reflection of high-income group consumption patterns, ultimately fostering equitable carbon pricing mechanisms.

When it comes to the impact of ranking methods on distributional effects, household employment status played a pivotal role. This research demonstrated that when employing expenditure as a ranking metric, many households showcasing disparities between income and expenditure undergo noticeable transitions in their quantiles. The "Other" employment category—encompassing retirees, students, and several other statuses and ranking third in terms of dataset volume—-stood out. This group showcased significant shifts across different ranking methods. Many within this category, such as voluntary non-workers, tended to display low-income but high expenditure patterns, reflecting higher potential lifelong earnings or savings. Given these dynamics, using expenditure as a rank criterion seems to be a more equitable approach for this demographic. This finding highlighted that, although there may not be a one-size-fits-all ranking method, tailoring the choice of ranking method to the specific demographic profile when formulating carbon pricing policies can lead to more balanced and equitable outcomes for society.

In prevailing literature, this study provided an enriched perspective on the multifaceted distributional implications of carbon pricing by integrating varied evaluation angles. The ever-increasing prominence of carbon pricing underlined the significance of this paper's findings. The Input-Output (IO) model this study had embraced, though providing a focused lens on immediate effects within the electricity-industry nexus, offered invaluable insights. Such a framework proved indispensable for countries without access to granular, industry-specific carbon data, enabling them to better discern the influence of carbon pricing on households. Even as this paper's findings were deeply rooted in Taiwan's context, they had broader implications, enriching global discussions on carbon pricing, especially in the domains of household categorization and employment dynamics.

The endeavor of this paper, while comprehensive, was not without its limitations. The SFIE dataset employed in this study left some high-carbon footprints, such as air travel and meat consumption, less defined. These ambiguities might present skewed effects across different income groups, magnifying the regressive nature of certain activities. The IO model's design primarily gave a short-term glimpse, sidelining potential shifts in consumer behavior in response to pricing changes or the long-term evolutionary trajectory of industries. Other considerations, like the potential for carbon pricing to recalibrate household incomes, as underscored by some studies, also factor into the broader picture. This paper's current framework, due to the lack of an exhaustive industry-specific carbon emission dataset, narrowed down indirect emission evaluations to electricity. While this provided a general overview, a more granular data set would allow for a deeper, country-specific understanding. As the realm of carbon pricing continues its dynamic evolution, this study and its simulation-oriented perspectives serve as a beacon for policymakers. The multi-faceted nature of carbon pricing necessitates a continuous cycle of research and dialogue tailored to diverse national contexts. Beyond just an income-centric focus, it is imperative to consider the sprawling impacts of carbon pricing across various household attributes, such as gender, age, and geographical location. This suggests the need for a multi-dimensional analytical approach, paving the way for promising avenues in future research initiatives.

CRediT authorship contribution statement

Cheng-Hsiang Shei: Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft. **John Chung-En Liu:** Conceptualization, Investigation, Supervision, Writing – review & editing. **I-Yun Lisa Hsieh:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2024.141446.

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